

Web Application Development

IS4102

Lecture Note 01 – Web Versions

K T Dananjali

Lecturer (Unconfirmed)

Department of Computing and Information Systems

Faculty of Computing

Assessment....

- Quizzes + Assignments – 40%
- Final Exam – 60%

Content

Web Application Architecture:, HTTP Web servers, Web Browsers, Dynamic Web Applications(architectures, clients, servers, dynamic content). Modeling Web Data: Data Model: Semi structured data, XML and ebXML Concept, Web Data Management with XML, XPath and XQuery, Typing of semi Structured Data, XML Query Evaluation, Managing an XML Database, Web Data Semantics and Integration: Ontologies, RDF, and OWL, Querying Data through Ontologies, Data Integration, Wrappers and Data Extraction with XSLT, Building Web Scale Applications: Web search Web Security

➤ Practical: Web applications development using PHP, ontology development

End of this lesson, you will be able to

- Distinguish between WWW and the Internet.
- List the advantages of WWW.
- Compare and contrast Web 1.0, Web 2.0, and Web 3.0
- Give the application of Web 2.0
- Distinguish between Server Side Dynamic Web Pages and Client Side Dynamic Web Pages.

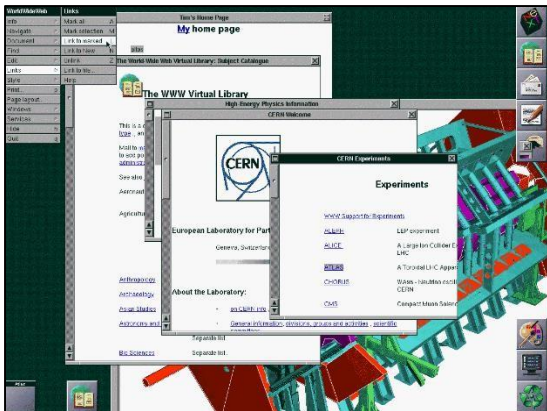
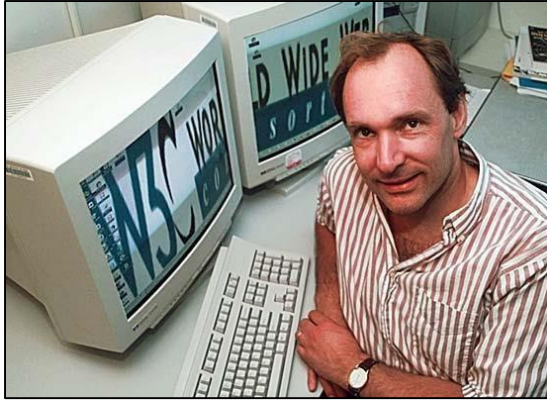
WWW (World Wide Web)

- The World Wide Web, or simply WWW, is a way of accessing information over the medium of the Internet.
- It is an information-sharing model that is built on top of the Internet.
- A worldwide collection of electronic documents. Each electronic document is called a Web page
- Can contain text, graphics, audio, video, and built-in connections

Advantage and Disadvantages of WWW

Advantages	Disadvantages
Easy access to information (Able to find any information: education, news, research instantly)	Security risks
Communication (Enables email, social media, chatting, and video calls)	False information
Online learning (Students can learn through websites, videos, and courses anytime)	Addiction
E-commerce (online shopping)	Privacy issues
Entertainment	Malware and viruses
Global connectivity	Cyberbullying

Tim Berners-Lee

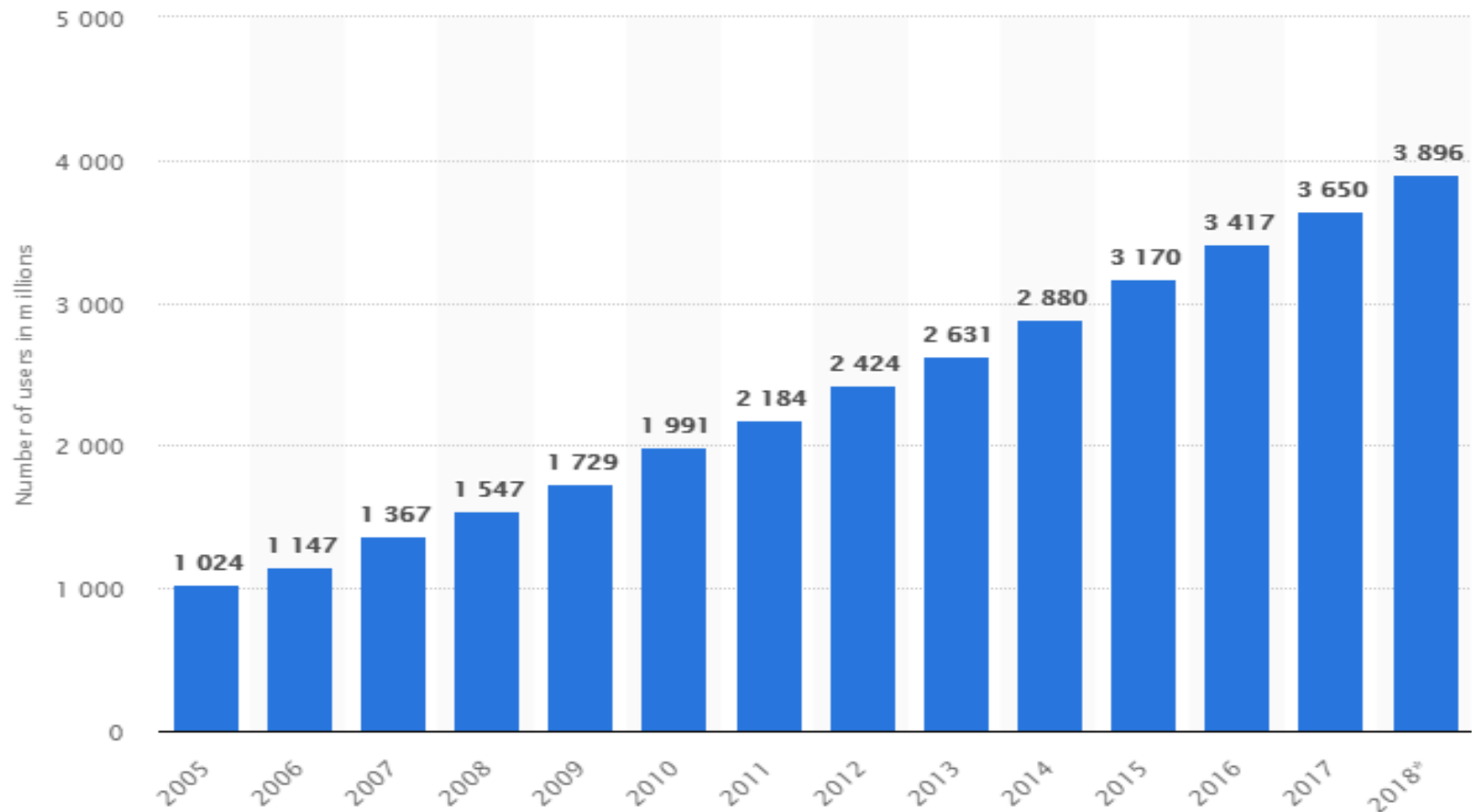


Tim Berners-Lee is shown here, along with the first picture posted on the Web and a screen shot from an early version of his Web browser.

Internet

- Largest network in the world that connects hundreds of thousands of individual networks all over the world.
- The development of the Internet began with early research networks like ARPANET, which was created in the late 1960s.
- Rather than moving through geographical space, it moves your ideas and information through cyberspace.
- The Internet provides many services, including the World Wide Web (WWW), email, file sharing, and online communication
- No one owns it.
- It has no formal management organization.

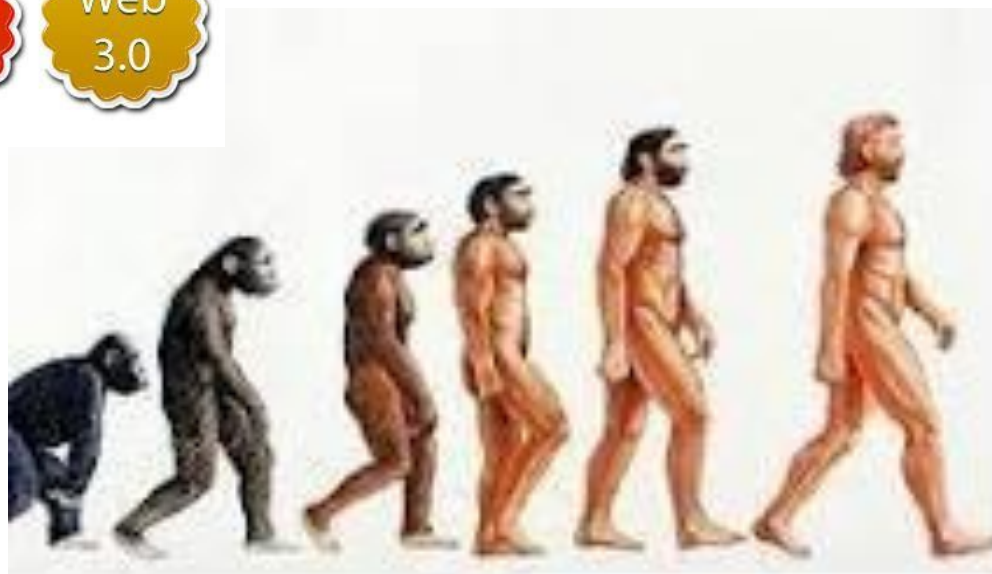
Number of Internet users



Number of Internet users

TOP 20 COUNTRIES WITH HIGHEST NUMBER OF INTERNET USERS - DECEMBER 31, 2017						
#	Country or Region	Population, 2018 Est.	Population 2000 Est.	Internet Users 31 Dec 2017	Internet Users 31 Dec 2000	Internet Growth
1	China	1,415,045,928	1,283,198,970	772,000,000	22,500,000	3,331 %
2	India	1,354,051,854	1,053,050,912	462,124,989	5,000,000	9,142 %
3	United States	326,766,748	281,982,778	312,322,257	95,354,000	227 %
4	Brazil	210,867,954	175,287,587	149,057,635	5,000,000	2,881 %
5	Indonesia	266,794,980	211,540,429	143,260,000	2,000,000	7,063 %
6	Japan	127,185,332	127,533,934	118,626,672	47,080,000	152 %
7	Russia	143,964,709	146,396,514	109,552,842	3,100,000	3,434 %
8	Nigeria	195,875,237	122,352,009	98,391,456	200,000	49,095 %
9	Mexico	130,759,074	101,719,673	85,000,000	2,712,400	3,033 %
10	Bangladesh	166,368,149	131,581,243	80,483,000	100,000	80,383 %
11	Germany	82,293,457	81,487,757	79,127,551	24,000,000	229 %
12	Philippines	106,512,074	77,991,569	67,000,000	2,000,000	3,250 %
13	Vietnam	96,491,146	80,285,562	64,000,000	200,000	31,900 %
14	United Kingdom	66,573,504	58,950,848	63,061,419	15,400,000	309 %
15	France	65,233,271	59,608,201	60,421,689	8,500,000	610 %
16	Thailand	69,183,173	62,958,021	57,000,000	2,300,000	2,378 %
17	Iran	82,011,735	66,131,854	56,700,000	250,000	22,580 %
18	Turkey	81,916,871	63,240,121	56,000,000	2,000,000	2,700 %
19	Italy	59,290,969	57,293,721	54,798,299	13,200,000	315 %
20	Egypt	99,375,741	69,905,988	48,211,493	450,000	10,613 %

Evolution of the Web



Web 1.0

Web 2.0



WEB 3.0



Web 1.0

- It was widely used between 1998 and 2001, and it is still used beside Web 2.0 in almost all websites.
- The main features of 'Web 1.0' were hyperlinking and bookmarking of the web pages.
- There was no flow or communication between the consumer and the producer of the information. It is one-way broadcasting.
- The best examples of static websites



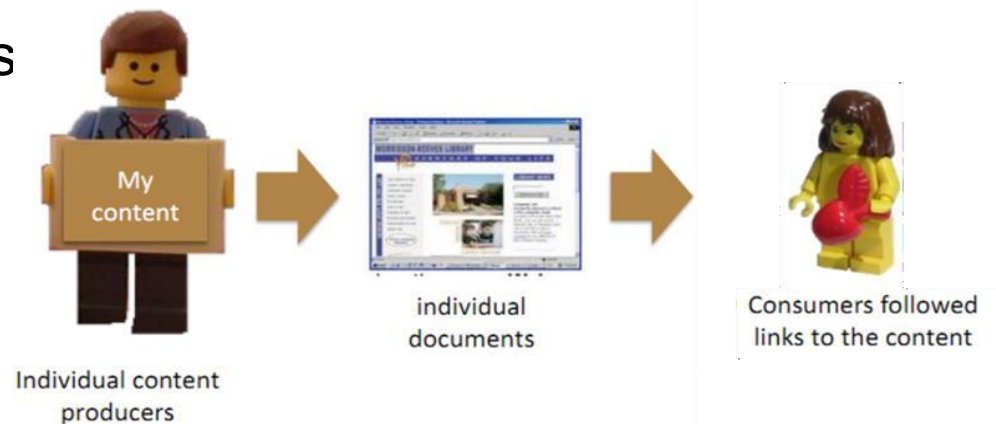
Web 1.0

Some of the technologies developed during this stage of the Web include:

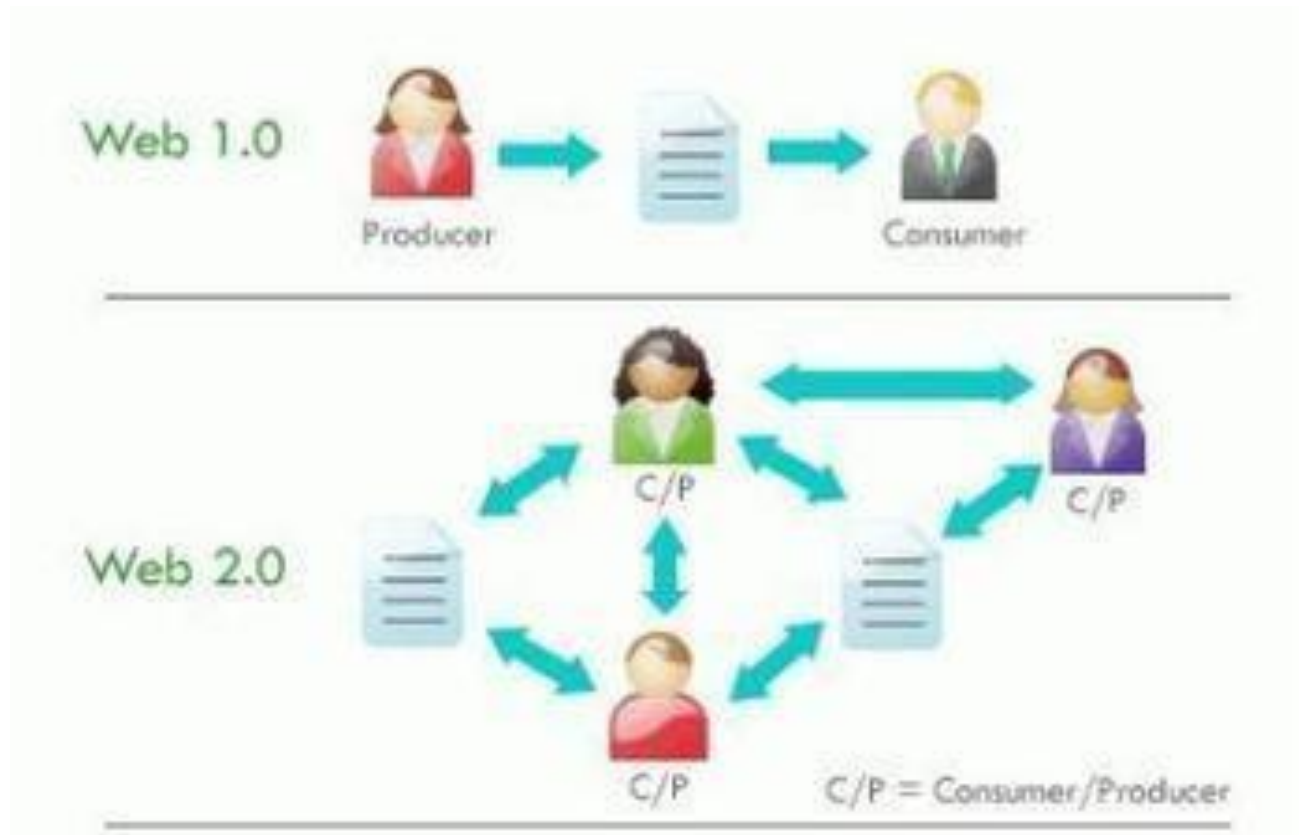
- File and Web Servers
- Content and Enterprise Portals
- Search Engines (AltaVista, Yahoo!)
- E-mail (Yahoo!, Hotmail)
- P2P File Sharing (Napster, BitTorrent)

Web 1.0 - Disadvantages

- Read-only Web
- Limited user interaction
- Keyword-based (dumb) search----- Web Directories
- The Lack of Standards



Web 2.0



The term was popularized by Tim O'Reilly and Dale Dougherty at the O'Reilly Media Web 2.0 Conference in late 2004,

Web 2.0

What is Web 2.0

Web 2.0 refers to a second generation of services available on the internet that let people collaborate and share information online. They often allow for mass publishing (web-based social software). The term may include blogs and wikis.

Web 2.0 Space



- Some of them are technological components (e.g., AJAX, RIA's, and XML/DHTML)
- Some are principles (e.g., participation, collective intelligence, and rich user experience)
- Some are applications and tools (e.g., Wikipedia, Flickr, and Mashups).

Web 2.0

- Community
 - Users organize themselves and work in partnership with common goals
- Active participation
 - Users move from passive role (reading) to active role (authoring)
- The Wisdom of Crowds:
 - Individual users add value
 - Aggregate data into a collective thought
 - Applications get better/smarter the more people use them

Web 2.0 Technologies and Applications

- User-Generated Content (e.g. Flickr and YouTube)
- Web Content Sharing (e.g. Digg)
- Social Bookmarking (e.g. Del.icio.us)
- Blogs
- Wikis
- AJAX

Web 2.0 Opportunities

- Collaborate more easily:

- Internally
- Externally



- Enable people to help each other – peer to peer collaboration
- Aggregate constituent wisdom: “The whole is smarter than the one”

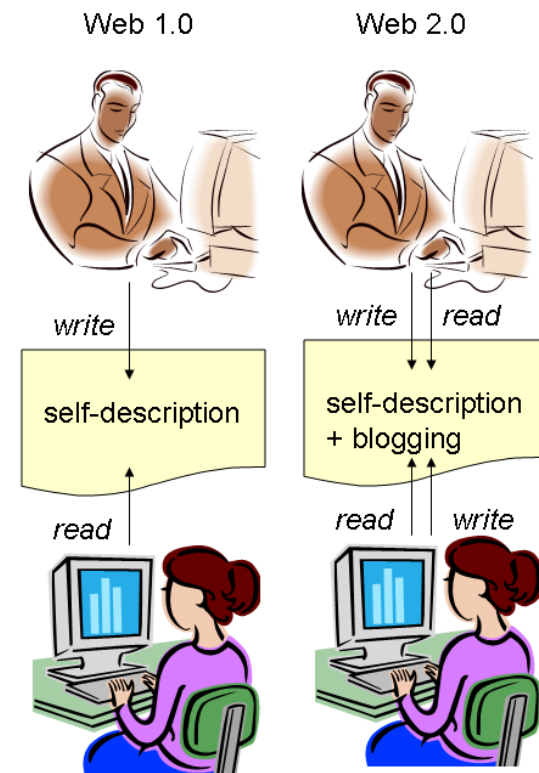
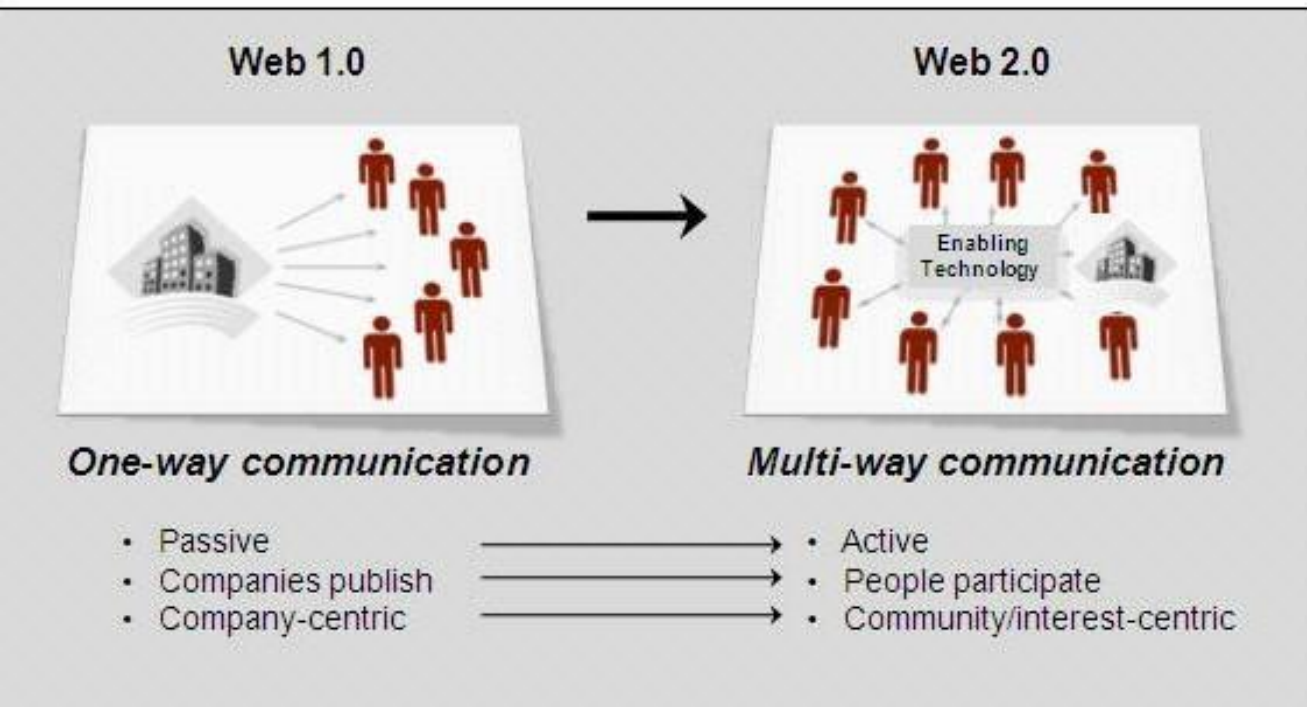
Web 2.0 Challenges

- Privacy issues
- Control of Content
- Anonymous postings (yes or no?)
- User Trust - can change the content of others
- Vandalism
- Plagiarism and Copyright Infringement

The Difference between Web 1.0 and 2.0

Feature	Web 1.0	Web 2.0
Authoring	personal web sites	blogging
Information sources	Britannica Online	Wikipedia
Content creation & maintenance	via CMS	via Wikis
Data storage	local disk	online disk
Online advertising	banners	Google AdSense
Online payment	bank account	PayPal

The Difference between Web 1.0 and 2.0



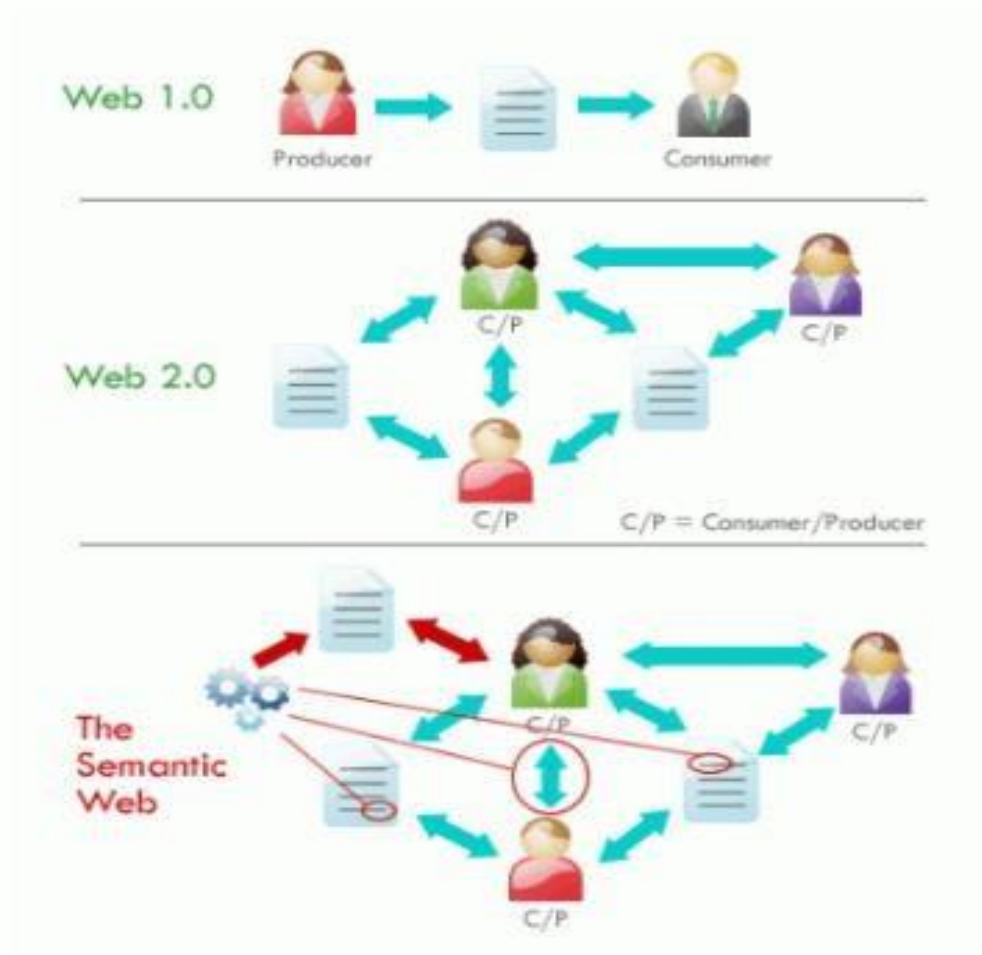
Web 3.0

This will be the INTELLIGENT

Web!

The Semantic Web!

Next...



Web 3.0

Machine Centric Web:

- Purpose: make the web readable by machines and not only by humans
- The third generation of Web technologies and services that emphasize a machine-facilitated understanding of information on the Web.

1994-2000

WEB 1.0
Information web
Weblearning
Reading 1.0

Mostly READ web.
Limited number of authors.
Million of users.
Static information and home pages.
HTML portals.
Web forms.
Directories (taxonomy).
Ex. Britannica Online.

2000-2010

WEB 2.0
Social web
Collaborative learning
Weblearning 2.0

Mostly READ-WRITE web
COMMUNICATIVE web.
Millions of authors
Billions of users.
Great activity (blogs, wikis stb.).
Sharing content.
XML, RSS.
Web applications.
Tagging („folksonomy“).
Ex. Wikipedia.

2010-2020

WEB 3.0
Semantic web
Personalized web
Learning 3.0

Mostly MOBILE and PERSONAL web.
Focus on individual user.
Integrated on-demand dynamic content.
Widgets, mashups, metadata, dynamic web services, ontology.
User behavior.
Active user engagement.
Ex. Apple Siri.

2020-2030

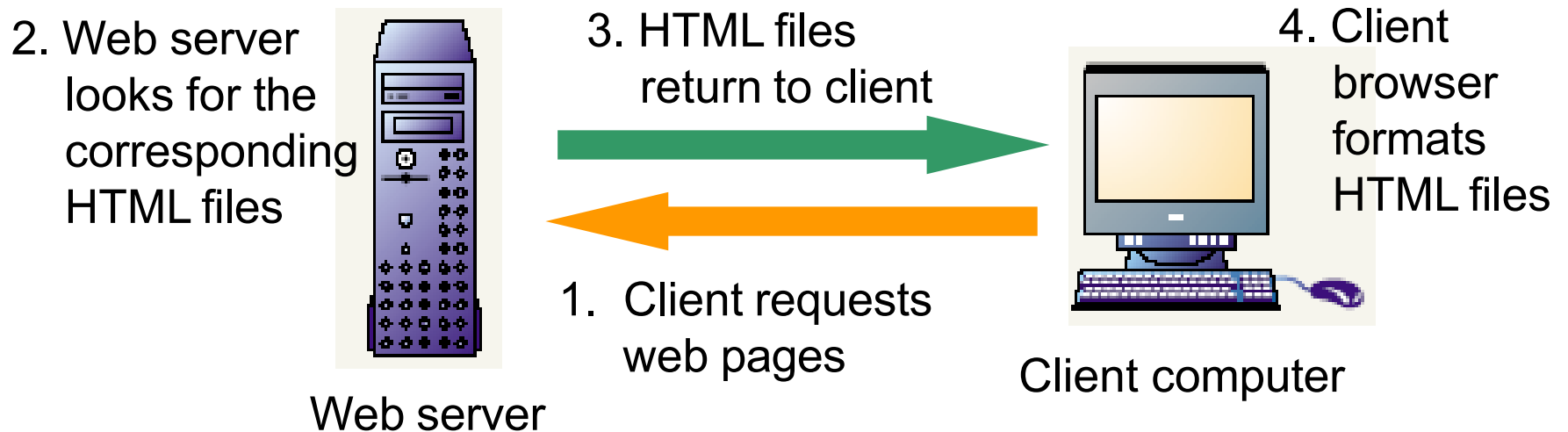
Mostly SELF-LEARNING, SELF-ORGANIZING web.
Focus on individual user + subject domain + level of knowledge.
Internet as a universal world computer.
Web as a universal knowledgebase.
Ex. Smart devices without screen.

Static and Dynamic Web Pages

Static Web Pages Vs Dynamic Web Pages

Static Web page

A Web page with contents that remain fixed and unchanged once the author has created it.



Static web page



A screenshot of a Notepad window titled 'hello_world - 記事本'. The window contains the following HTML code:

```
<html>
<head>
<title>Hello world</title>
</head>

<body>
<h2><font face="Arial">Hello world!</font></h2>

</body>
</html>
```



A static document is simply any web page that is saved to disk and passed back to a requesting browser without changes.

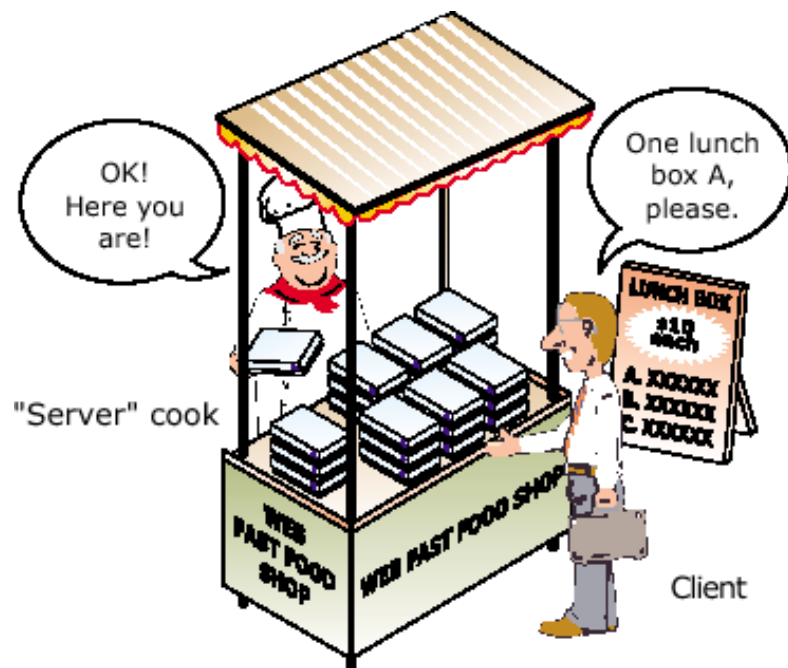
Dynamic Web page

A Web page with contents generated “on-the-fly” by the server or the client computer according to the client’s requests.

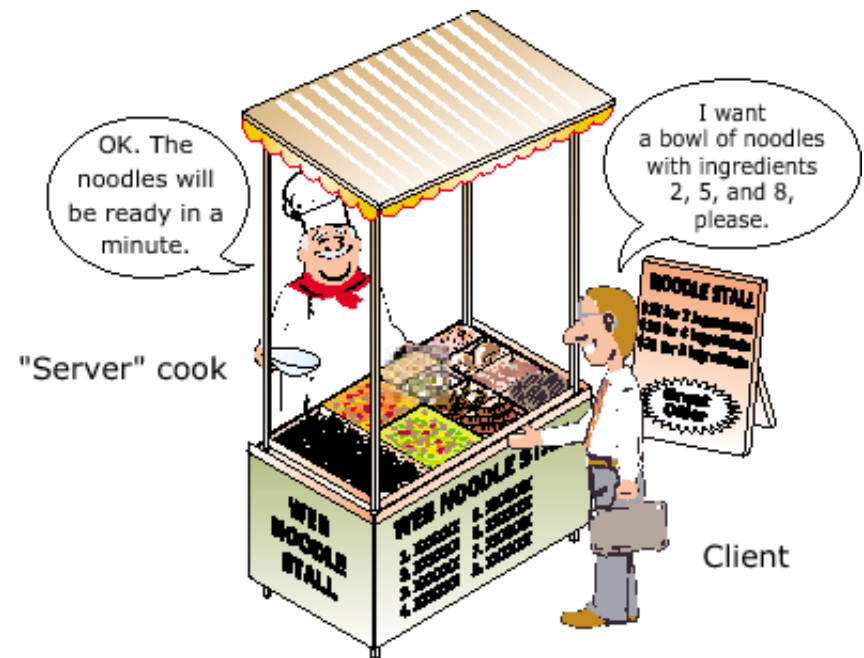
e.g. return the system’s current date and time to the surfer browser (written in Active Server Page).

A dynamic Web page is any Web page whose content is changed by a program or script at the time the page is requested.

Static Web Pages Vs Dynamic Web Pages



Buying ready-made lunch boxes:
browsing static Web pages



Ordering noodles in a noodle stall:
browsing dynamic Web pages

Client-Side Dynamic Web Pages Vs Server-Side Dynamic Web Pages

Classification of dynamic Web pages:

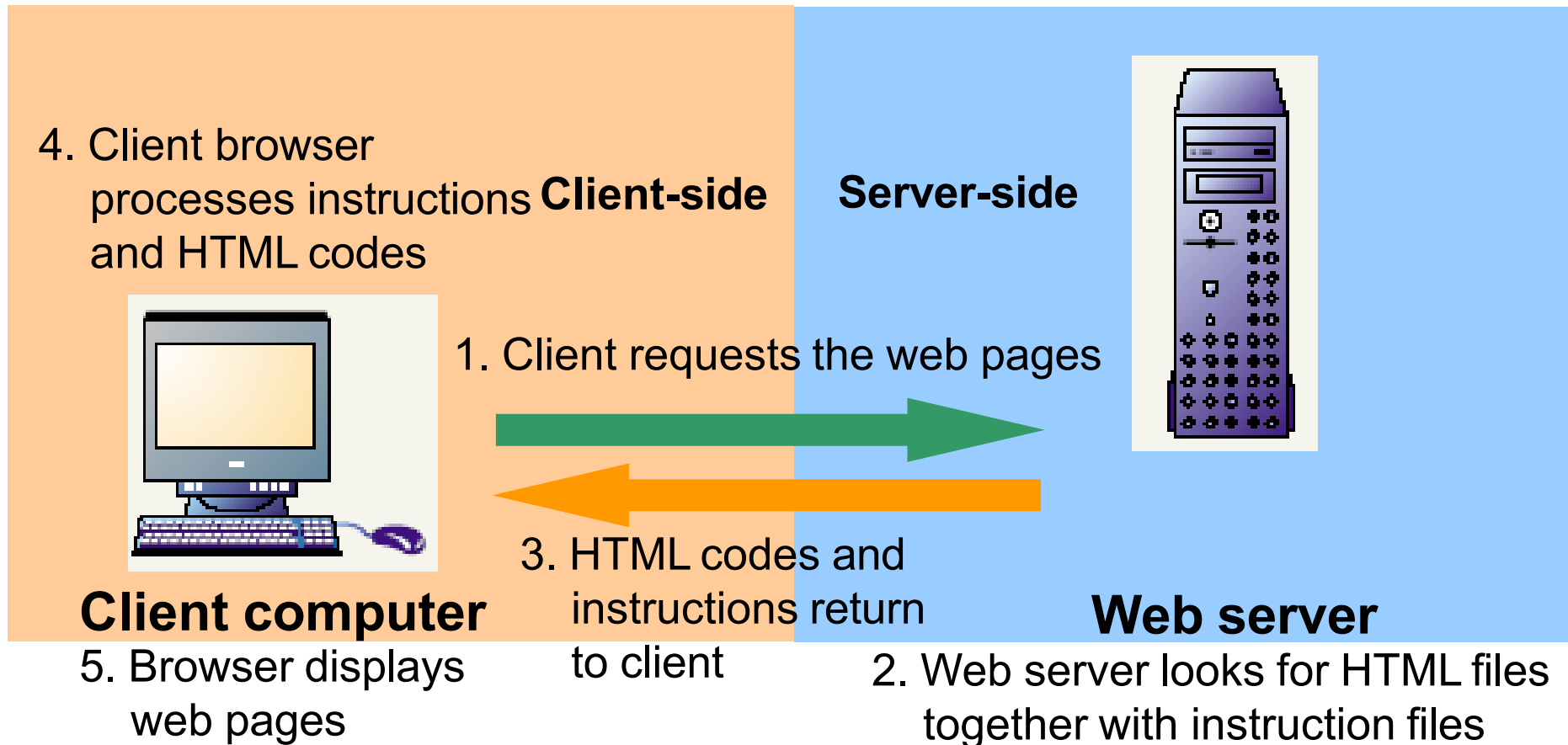
Client-side dynamic Web pages

- The creation of the Web pages takes place in the client computer.

Server-side dynamic Web pages

- The server is responsible for creating the Web pages.

Client-side dynamic Web pages



Client-Side Scripting

- Browsers can fetch certain scripts (client-side scripts) or programs along with documents, and execute them in “safe mode” at the client site
 - JavaScript
 - Macromedia Flash and Shockwave for animation/games
 - Applets
- Client-side scripts/programs allow documents to be active
 - animation by executing programs at the local site
 - ensure that values entered by users satisfy some correctness checks
 - Permit flexible interaction with the user.
 - Executing programs at the client site speeds up interaction by avoiding many round trips to server

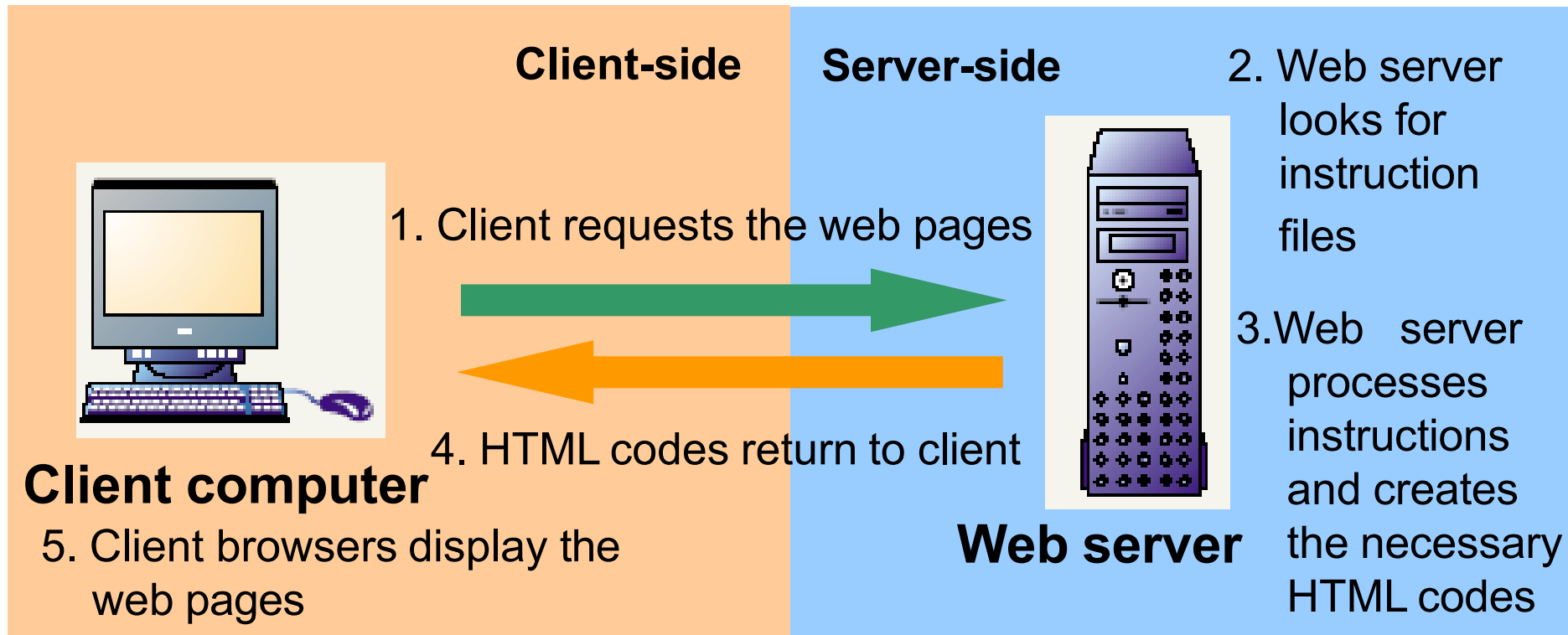
Client-Side Scripting and Security

- Security mechanisms needed to ensure that malicious scripts do not cause damage to the client's machine
 - Easy for limited capability scripting languages, harder for general-purpose programming languages like Java

E.g. Java's security system ensures that the Java applet code does not make any system calls directly

- Disallows dangerous actions such as file writes
- Notifies the user about potentially dangerous actions, and allows the option to abort the program or to continue execution.

Server-side dynamic Web pages



Q1. Compare the Client-Side Dynamic Web Pages and Server-Side Dynamic Web Pages.

Client-Side Dynamic Web Pages Vs Server-Side Dynamic Web Pages

	Client-Side Dynamic Web Page	Server-Side Dynamic Web Page
Instruction codes revealed to user	Yes	No
Extra time required to download instruction codes in addition to HTML codes	Yes	No
Need to cater for browser differences	Yes	No
Accessibility to server resources (such as a database)	Low	High

Comparison between client-side and server-side dynamic Web pages

Which Should I Use? Client- or Server-Side?

- If you want to have dynamic client forms with client-side validation, you must use client-side scripting.
- If you want your site to have highly interactive pages, you should use client-side scripting.
- If you need to provide your client with advanced functionality that can be created only using ActiveX controls, you must use client-side scripting.
- If you want to protect your source code, you must use only server-side scripting. All client-side source code is transferred to the browser.

Which Should I Use? Client- or Server-Side?

- If you need to track user information across several Web pages to create a "Web application," you must use server-side scripting.
- If you need to interact with server-side databases, you must use server-side scripting.
- If you need to use HTTP server variables or check the capabilities of the user's browser, you must use server-side scripting.

	JavaScript	VBScript	Java Applets	ASP	JSP	CGI	PHP
Processing takes place in	Client-side	Client-side	Client-side	Server-side	Server-side	Server-side	Server-side
Browser support (Netscape, I.E.)	Both	I.E. only	Both	Both	Both	Both	Both
Intruction codes disclosed to users?	Yes	Yes	No	No	No	No	No
Need to download instruction files in addition to HTML codes?	No	No	Yes (Applet)	No	No	No	No
Scripting or full-fledged programming language	Scripting only	Scripting only	Full prog. lang.	Scripting only	Full prog. lang.	Full prog. lang.	Full prog. lang.

Comparison between Client-Side and Server-Side Processing Technologies